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An Optimized Proxy Modeling Framework for Simultaneous Well Placement Across Multiple Development Units

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Abstract

This paper presents an accelerated and more granular version of the scalable proxy modeling framework for simultaneous well placement optimization by introducing asynchronous workflows and including multiple development units (DUs) scenarios. Traditional simulation-based well placement is computationally intensive for large reservoirs, motivating the use of AI-based proxy models for rapid optimization. Our methodology integrates DU-specific parameters with parallelized, vectorized workflows to evaluate multiple scenarios simultaneously. The framework was applied to an ADNOC carbonate reservoir which is divided into two major and fifteen minor units, achieving a 70× speed-up over conventional approaches while improving NPV by 4.6%. Enhanced trajectory control, orientation awareness, grid interpolation, and DU-aware zoning ensure operational feasibility and scalability. The novelty of this work lies in simultaneously optimizing across multiple reservoir compartments with a demonstrated multifold computational speed-up, a challenge rarely addressed in prior studies. The framework is scalable to both greenfield developments and brownfield projects, making it widely applicable across diverse carbonate settings. Results validate the framework's effectiveness for complex, compartmentalized fields. Future work includes integrating multi objective optimization (e.g., production profile plateau constraints) to further broaden its applicability.

Introduction

Optimizing the placement of wells is a critical challenge in reservoir development and management. Identifying optimal locations for new producers and injectors can maximize oil recovery and net present value (NPV), but this requires evaluating numerous scenarios due to the complex geology and dynamic flow behavior of reservoirs [1,2,3]. Traditionally, full-physics reservoir simulations are run for various well configurations to find the best outcome, which is extremely computationally intensive. For large, complex fields, exhaustive simulation-based search can take hours to days per scenario [4], making it impractical to thoroughly explore the well placement design space with brute-force methods. In practice, this often forces engineers to test only a limited set of development plans, potentially missing the true optimum. The

high stakes of drilling costs and long project timelines mean that any suboptimal well placement can have significant economic repercussions, further underscoring the need for efficient optimization techniques.

This challenge has motivated the development of proxy models as faster approximate simulators. These proxies act as "surrogate reservoir models", capturing the key input–output relationships of the reservoir in a fraction of the time [5, 12], using statistical or machine-learning techniques, such proxies mimic the simulator's results within acceptable error tolerances, dramatically accelerating the optimization process. By training on data from a handful of simulated scenarios, a proxy can learn to predict outcomes (like production or NPV) for new well configurations almost instantly, whereas a full simulation might take hours. Recent reviews have highlighted that data-driven proxies (e.g., artificial neural networks, Gaussian processes) and reduced-order models are increasingly central in well placement optimization [2, 12]. These approaches allow engineers to evaluate many more feasible development scenarios than with direct simulations.

Another complicated factor in field development is the presence of multiple development units (DU) or flow compartments within a single reservoir [6, 7, 13]. Giant carbonate reservoirs in the Middle East, contain hidden baffles and barriers that partition the field into semi-independent units. Each unit (or flow zone) can have distinct rock and fluid properties, meaning that an optimal well pattern in one unit may not be optimal in another. Past studies emphasize the need for optimization algorithms that account for such non-communicating flow units, ensuring effective drainage across all reservoir zones [8,9] extended well placement optimization to handle multiple flow units by selectively completing wells in different zones, enabling production from the entire reservoir while controlling unwanted water or gas production [9].

However, optimizing across multiple units simultaneously (rather than sequentially) introduces additional complexity, as it requires a unified framework to balance the well locations and allocations across the whole field. A sequential approach—optimizing one compartment at a time—might yield suboptimal overall results because it could over-invest in one unit while neglecting another. This complexity also increases the computational burden along with increased time requirement [1]. Therefore, a comprehensive strategy is needed that can simultaneously handle multi-unit interactions without becoming intractable.

Recent works on parallel computing and vectorization have shown that these techniques can significantly accelerate reservoir simulation and optimization workflows. By leveraging parallel processing via engaging multi-core processors simultaneously and using approximate matrix-based interpolations, studies have demonstrated order-of-magnitude speed-ups in simulation throughput [1,10,11]. The integration of parallel computing into the optimization framework ensures that even complex, high-dimensional problems can be solved within practical timeframes, making it feasible to incorporate more geological detail and more candidate solutions in the optimization process.

By applying a simultaneous multiple DU optimization, engineers can ensure an optimal distribution of wells among ADNOC Reservoir's unit, avoiding the risk of underdeveloped zones that might result from optimizing each unit in isolation. Moreover, the reservoir's sheer scale and well count make it an ideal candidate for parallel computing and vectorization during the proxy modeling and AI-based optimization frameworks – even small efficiency gains can translate into significant incremental NPV when scaled across such a large field [14].

In this work, we present a unified framework that integrates multiple DUs inclusion within the machine-learning proxy modeling with well placement optimization. We leverage parallel computing and vectorized placement engines to tackle computational challenges. Tasks are executed in parallel across DUs and realizations, dramatically shortening the time to convergence. The following sections describe the methodology – including proxy model selection, workflow design, and computational enhancements – and then present results from applying the framework to the ADNOC reservoir.

Methodology

We built on top of previously developed methods that includes creating proxies for large scale reservoir simulators [1]. The Proxy model evaluation is performed using the dedicated evaluation metric Top-K accuracy and robustness, which measures the models around recurrence, along with the commonly used mean squared error and R2 values. Top-k accuracy is a machine learning metric for multi-class classification that measures the percentage of times the true class label is among the top k predicted classes by the model. Here we measure the overlap of discounted NPV in the simulations and proxy predictions among top k cases. Top-k robustness for measure 1 denotes whether top 1 discounted NPV from simulations is within top k that from predictions by proxy model. Then using advance optimization schemes to optimize further based on the constraints and the chosen economic targets. The optimized field control parameter results are then fed to the well placement workflow to find optimized well locations for injectors and producers corresponding to optimal integrated cumulative production, injection volume. Within the well placement module, we are performing a sequence of steps involving creation of opportunity maps, zone isolation, well count analysis, final well placements and simulation validation [1,4,5]. Through this pipeline, the proxy model drastically reduces the time per evaluation, the optimization algorithm intelligently searches the design space, and the well placement module translates global optima into concrete drilling plans. Figure-1 describes how within the well placement module parallelization of workflow was achieved. To address the challenges posed by non-uniform spatial resolution in the X and Y axes of the reservoir grid, an algorithm was developed to compute an optimal grid spacing that accurately captures the highest frequency variations in the input data covering variations in both the X and Y direction. This enabled interpolation of reservoir properties onto a uniform grid, facilitating subsequent computations. An interpolation scheme was then implemented to estimate property values at newly generated coordinate points during optimization. Additionally, an orientation control feature was introduced, allowing the placement of wells at user-defined azimuths to reflect operational preferences and reservoir flow dynamics. This required the development of a modified masking algorithm that accounts for the directional extent of the wellbore and applies an additional buffer zone to mitigate potential well interference and collision risks.

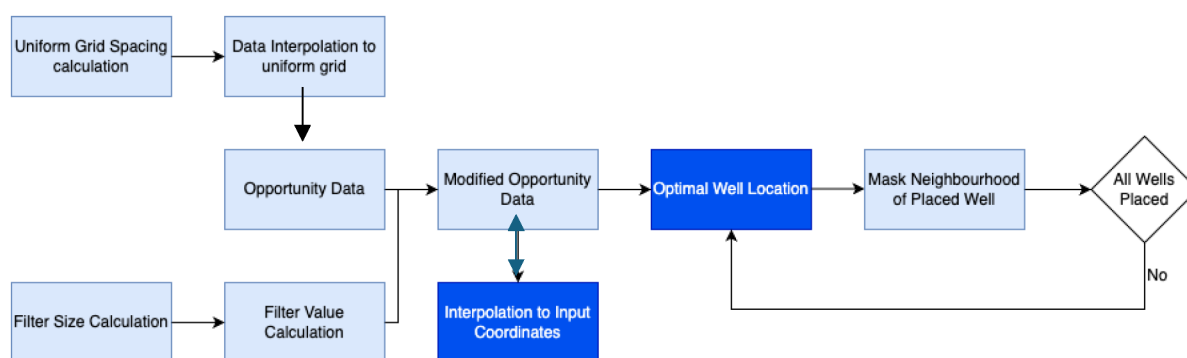


Figure 1—Modified pipeline for well placement to incorporate parallelization and scalability.

Historically, these operations—including neighborhood masking and opportunity map updates—were performed iteratively for each candidate well, resulting in considerable computational overhead [1]. In the revised workflow, these steps were vectorized and executed using custom matrices derived from the input parameters. This parallelized approach enables simultaneous evaluation of multiple candidate locations and orientations, resulting in a substantial reduction in computational time while preserving accuracy. The enhanced speed and scalability of this methodology are particularly advantageous for optimizing well placement across multiple DUs in large, heterogeneous reservoirs, as demonstrated in the results section.

To incorporate the dynamics of the ADNOC reservoir into consideration, it was partitioned into primary DUs, namely Unit-A and Unit-B, each characterized by distinct well spacing parameters (dX , dY),

production and injection lengths, and toe-to-heel values. To further enhance spatial resolution and account for reservoir heterogeneity, these primary units were subdivided into minor-units, with each minor-unit assigned a unique well azimuth to reflect local geological trends and operational constraints. The schematic of minor units and the major units are demonstrated in figure-2 along with an explanation of the various well placement parameters.

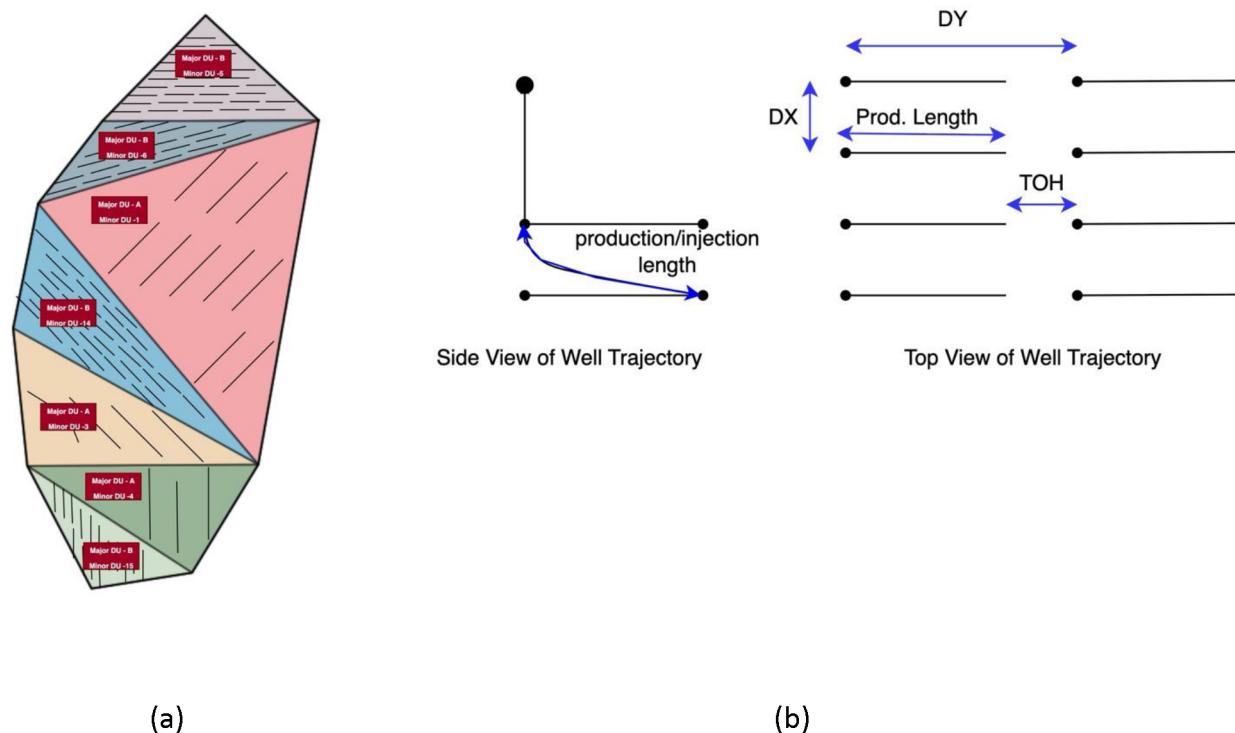


Figure 2—Schematic representation of DUs, based on lithology, permeability, and porosity characteristics, illustrating the need for unit-specific well design strategies. The figure also illustrates variations in production length, spacing (DX, DY), and trajectory orientation (TOH) across DU, as shown above in figure(b), illustrating how varying these parameters can lead to different well placement outcome.

An optimized well placement algorithm was executed independently for every combination of major DU and its associated minor-units. This process allowed for tailored placement strategies that respect the geospatial boundaries and directional preferences of each zone. High-ranking wells from each optimization run were systematically mapped and assessed for compliance with minor-unit and DU boundaries. Finally, the output from all combinations were merged to generate an integrated, field-wide well placement plan, which was subsequently verified using full-physics reservoir simulation to ensure operational feasibility and production targets.

Application in ADNOC carbonate reservoir

The proposed workflow was applied to a large ADNOC carbonate reservoir consisting of two major DUs and fifteen minor DUs, which are divided based on rock properties, such as lithology, porosity and permeability etc. [13]. The framework successfully optimized well placement parameters across these units, generating a unified well placement plan. Our proposed architecture has unified the modelling of 15 different well parameter combinations into a single structure thus, proving the pathway for scalability across even large fields with more heterogeneous properties and varying operational constraints. Similar benefits were achieved with its integrated trajectory management with appropriate orientations per combination ensuring simulation-ready well configurations. This was performed keeping into considering the large size of the reservoir with over 10 million active grids, with runtime of over 14 hours with 128 CPUs per case. The

number of producers and injectors were taken to accommodate the remaining oil in place keeping in mind the available porosity and permeability. In this application, we have explored two different operational limits which are C1 and C2. C1 represents a condition with a higher facility capacity and C2 represents lower facility capacity due to lower budget.

The proposed approach demonstrated substantial computational gains by its vectorized implementation of optimization loops achieving approximately 70× speed-up over previously deployed conventional methods, details of the time gain are mentioned in [table 1](#). The Experiment was repeated by changing the orientation, spacing and production/injection length combinations to ensure that the results are robust and data agnostic. This is demonstrated by similar results from both the trials (1,2) added in [Table-1](#). Adding to that, the architecture enabled simultaneous implementation of well placement for individual DU and SU combination, led to a speed up of 15 folds as compared to the sequential approach of treating every combination of DU, SU as a new independent case.

Table 1—Comparison of runtime and iterations before and after of vectorization

Trial	Without vectorization	With vectorization
1	23mins	19 Seconds
2	23.5mins	19 Seconds

Based on the workflow describe in the methodology section, we carried out the proxy model creation and its evaluation to allow its usage in the optimization step. As described in [figure 3](#) and [table 2](#), R2 score of the economic model is high (0.97), almost all yes in the Top K robustness with a high score in the Top K accuracy indicate meta model-economic is representative.

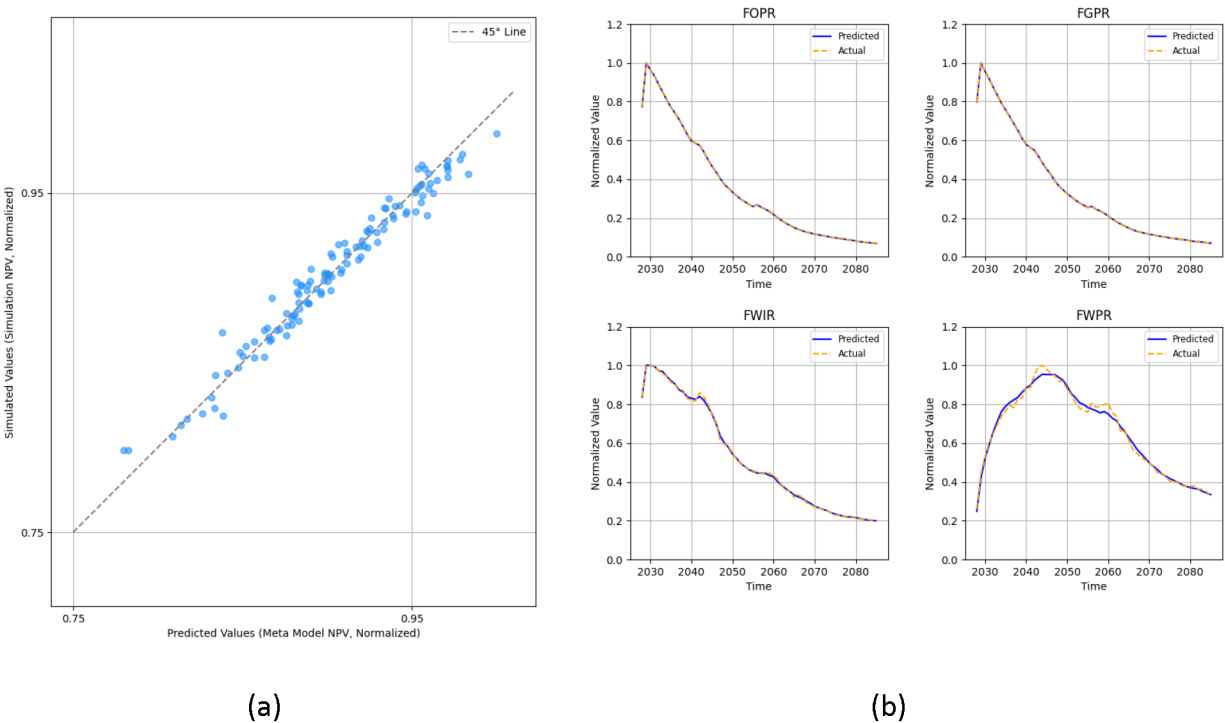


Figure 3—Comparison plot demonstrating the accuracy of the meta-model in predicting both economic metrics (Discounted NPV) and production profiles (Oil, Gas, Water) using single-time model.

Table 2—Top-K results referenced in methodology section, highlighting the robustness and accuracy of the model for different thresholds.

Top K Robustness						
Measure	Single time Model			Economic Model		
1	Yes	Yes	Yes	Yes	No	Yes
3	Yes	Yes	Yes	Yes	Yes	Yes
5	Yes	Yes	Yes	Yes	Yes	Yes
10	Yes	Yes	Yes	Yes	Yes	Yes
Top K Accuracy						
Measure	Single time Model			Economic Model		
1	100	100	100	100	0	100
3	100	66	66	100	33	66
5	100	60	80	100	40	80
10	100	70	90	100	90	90

The framework achieved marked improvements in key economic and production metrics. Notably, Forecasted Production increased by 5.8% compared to baseline and Discounted Net Present Value (NPV) achieved an uplift of 4.6% over the existing baseline. The improvement was validated using the simulation run as demonstrated in [figure 5\(a\)](#) and [6\(a\)](#). The improvement in economic metrics comes along with the notable fact that there was no significant improvement in the number of optimized wells.

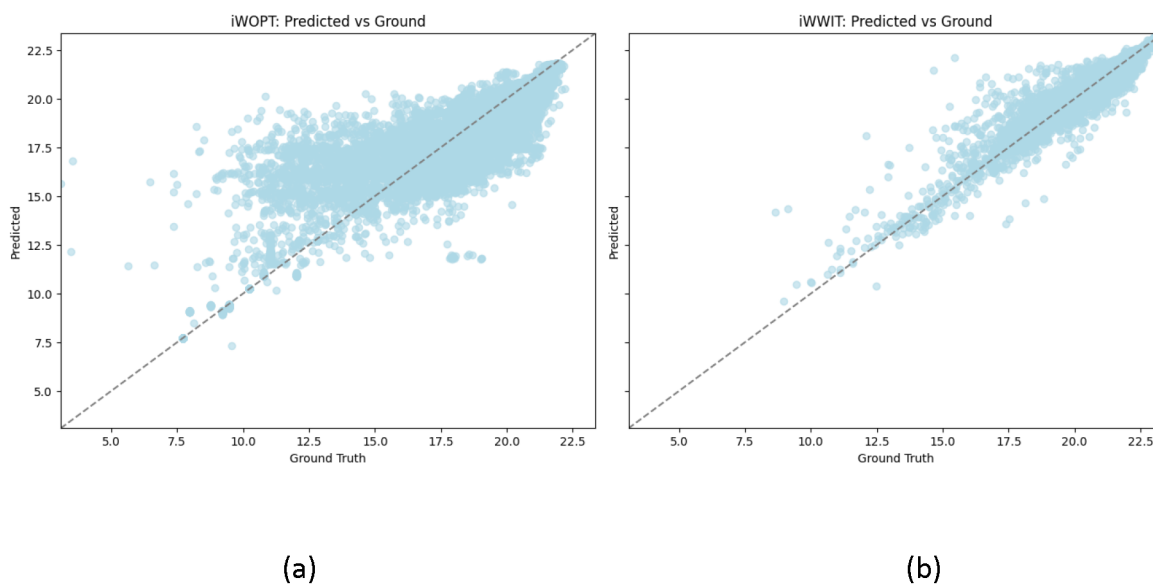


Figure 4—Model performance evaluation for cumulative well-level production: WOPT with $R^2 = 0.804$ and WWIT with $R^2 = 0.905$, indicating strong predictive accuracy.

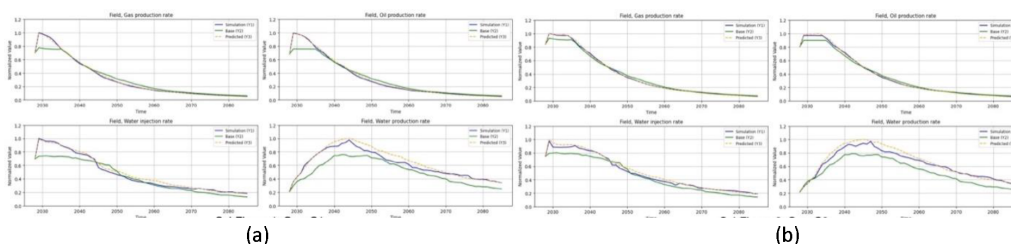


Figure 5—Validation of simulator results at the field level for (a) basic configuration (C1) and (b) refined configuration (C2), demonstrating performance consistency even in refined constraints. The comparison covers multiple Gas, Oil and Water production/inject

Benefit of this approach and the possibility of running well placement again quick on another set of values. Which in this case is C2. In our scenario we modified the range for rate of oil production, water injection and well oil production (multiplier). This was done to find a more practically suited scenario that meets the existing set of requirements. The new constraints although narrowed but still showed a 5.7% economic enhancement over the baseline. Simulation compared to base model and application prediction are described in figure 5(b) and 6(b).

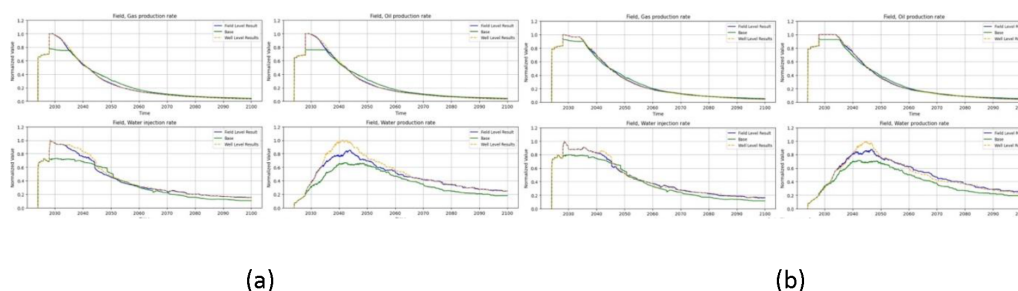


Figure 6—Well-level verification of simulation outputs for (a) basic (C1) and (b) refined (C2) cases, demonstrating performance consistency even in refined constraints. The comparison covers multiple Gas, Oil and Water production/injection metrics over

This results section highlights the transformative potential of the workflow in accelerating well placement creation and optimizing reservoir performance under complex operational constraints. Since a proxy model has been developed, it only takes fractional time to complete the optimization process when the constraints boundary conditions change, which gives us the flexibility to explore possibilities under various facility budgets and plans.

Conclusion

This study presents an integrated, scalable framework for simultaneous well placement optimization across multiple DUs leveraging the capabilities of machine learning proxies. Engineering enhancements of parallel computation and vectorized workflows had been added to the pipeline to speed up the computation efficiency. With the framework, we effectively reduced simulation costs while maintaining high accuracy in predicting production and economic outcomes.

Application to the ADNOC reservoir demonstrated substantial improvements compared to conventional way of well placement: a 5.8% uplift in production, a 4.6% increase in NPV, and a 70× reduction in computational runtime. The division of the reservoir into major and minor units ensured the balance between geologically consistence and operationally feasibility of well placements, while parallelization further increased the computation efficiency of evaluation of complex scenarios across multiple minor units within the major units.

This framework holds promise for broader deployment in other large-scale, compartmentalized reservoirs where one single simulation model has been established with unstructured partitions of regions that are

used for well placement. It can optimize parameters for each development unit in one go without costing more on computation compared to optimizing sequentially for each unit. Future enhancements will explore incorporating water injection and plateau-length constraints directly into the optimization objective, further increasing its realism and value to field development planning.

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